

Device-Wise Use Cases

Awesom Living Smart Home Platform — Complete Audit

Generated: September 18, 2026 | Covers all development sessions (April 2025 — September 2026)

82	9	82
Total Use Cases	Device Categories	Bugs Fixed / Features

Device	Use Cases
GLK Vital Tracker	18
Motion Sensor	6
Presence Sensor	2
Door/Window Contact Sensors	5
Emergency Button	2
AltumView Sentinare	1
CP Plus Camera	6
Raspberry Pi Hub	22
Cross-Device / Platform	20

1. GLK Vital Tracker — Sleep / Vitals Sensor (18)

Sole vitals and sleep monitoring device. Replaced Emfit sensor.

- **Sleep date attribution fix** — GLK reports data at 2-6 AM; fixed logic so sessions attribute to the correct sleep date, not the calendar date of the raw timestamp
- **Snoring / activity / restless data leaking from previous day** — data from a prior night was bleeding into the current day's response; fixed fetch window boundaries
- **ROOT CAUSE of empty sleep sessions** — GLK source detection missed cloud webhook format (glk_cloud vs glk_local); sessions appeared empty because ingestion didn't recognize the payload
- **previous_sleep_session computation fix** — the 'compared to last night' comparison was pulling the wrong session
- **GLK fetch window fix** — API query window wasn't aligned with actual data arrival pattern (2-6 AM next day)
- **15-minute minimum duration filter** — added to discard spurious micro-sessions (sensor noise, brief power events)
- **Device-offline close in build_sleep_sessions** — sessions left open when device went offline are now properly closed
- **merge_close_sessions widened 15 min to 30 min** — adjacent sessions separated by brief gaps (bathroom trip) merged into one continuous night
- **110-minute data gap analysis** — analyzed raw data and proved the gap was GLK sensor going silent (hardware), not a software bug
- **device_id: null in create_emfit_log** — log creation wasn't attaching the device ID; fixed
- **Backfilled device_id on all 734 existing emfit_logs records** — retroactive data integrity fix
- **get_sleep_date_range window fix** — date range query for sleep data was off by one day
- **Only last sleep session in APIs** — enforced user requirement that APIs return only the most recent session
- **Removed _debug / _session_debug from API response** — cleanup of temporary debugging data from production response
- **GLK vitals completeness fix (v20)** — signal_quality field was extracted but never returned; fixed to forward all 13 raw device fields including heart_rate, respiration_rate, battery_level, signal_quality, etc.
- **GLK BLE transport fix (v9)** — provisioning was failing (wifi_ack=false, 502) because BLE write used response=True which GLK fff1 characteristic doesn't support; fixed to response=False with 0.12s inter-chunk delay
- **GLK pairing 502 fix (v8)** — server.js reads glk_provision.py from wrong path; setup script only deployed improved version to one location. Fixed: copy into both locations
- **GLK emergency frame forwarding (v20)** — emergency (0x0D) frames now forward decoded fields (life_abnormality=true, status_code=5) instead of raw hex, enabling backend push notifications

Note: Resident dropdown kept for GLK pairing only (removed from all other device forms).

2. Motion Sensor — SONOFF SNZB-03 Zigbee (6)

Room-level motion detection for occupancy and 'no motion' alerts.

- **Motion sensor flickering + stale presence bug fix** — dashboard showed rapid on/off flickering and stale 'motion detected' that never cleared
- **Motion sensor pairing UI** — added 'Paired Motion Sensor' and 'Paired Window/Door Sensor' dropdowns to dashboard device form for door-aware occupancy linking
- **'Too long in room' no-motion alerts — 3 critical bugs fixed:** (1) Alert timer not resetting on new motion events, (2) Grace period calculation wrong, (3) Alert firing during normal occupancy patterns
- **Motion sensor alert grace period bug** — configurable grace period before 'no motion' alert wasn't being respected
- **Motion sensor alerts not firing — 3 root causes:** (1) Zigbee event path not triggering alert evaluation, (2) Home-based device lookup failing, (3) Alert threshold comparison inverted
- **Multiple paired window sensor filter fixes** — filter associating a contact sensor with its paired motion sensor had edge cases where wrong sensor matched

3. Presence Sensor — SONOFF SNZB-06P Zigbee (2)

mmWave-based presence detection — detects stationary persons (sitting/sleeping).

- **Occupancy showing false while person is sitting still** — presence sensor correctly reported occupancy but dashboard showed 'unoccupied' due to logic error in state aggregation
- **Door-aware occupancy logic (complete implementation)** — presence + motion + door sensor data combined: if door opens and presence drops, person left; if presence stays, person is still there despite door activity

4. Door/Window Contact Sensors — SONOFF SNZB-04 Zigbee (5)

Open/close detection for doors and windows. Can be paired with motion sensors.

- **Contact sensor filter fixed to use `normalize_sensor_type`** — sensor type strings weren't normalized, causing filter mismatches
- **Contact sensor filter updated: only hide those paired with motion sensor** — standalone door sensors appear in user app; only those paired as 'window sensor' hidden
- **Cloud backend: hide contact sensors from user app listing** — paired contact sensors are infrastructure (motion sensor helper), not user-facing
- **Paired window sensor event leaking to standalone Door Sensor** — when a contact sensor was paired with motion sensor, its events appeared on unrelated standalone door sensor log
- **Resident dropdown removed from contact sensor pairing form** — contact sensors don't need resident assignment

5. Emergency Button — SONOFF SNZB-01 Zigbee (2)

One-press SOS alert for the elderly resident.

- **Emergency button alert_log not saving to DB** — button press events were received via Zigbee/MQTT but alert log entry wasn't being persisted to MongoDB
- **Diagnosed empty alert_logs for emergency button** — traced full event path from Zigbee2MQTT to MQTT bridge to backend webhook to DB write to locate the break point

6. AltumView Sentinare — Fall Detection Camera (1)

AI-powered fall detection ONLY. Does NOT do presence, motion, or sleep.

- **"Danger - UNKNOWN" alert fix** — fall detection alerts showed "Danger - UNKNOWN" instead of proper alert message; fixed alert type mapping on both qa and master branches

Critical correction: AltumView does FALL DETECTION ONLY. AltumView provisioning requires WPA2 Personal WiFi.

7. CP Plus Camera — RTSP Video Stream (6)

IP camera for live video feed via RTSP, proxied through go2rtc on the Pi.

- **Camera stream debugging — COMPLETED** — full end-to-end troubleshooting of RTSP to go2rtc to WebRTC/HLS pipeline on the Pi
- **Camera checker CAMERA_ONLINE/OFFLINE resident lookups fixed** — periodic camera health check was failing to find associated resident, so online/offline status wasn't reported
- **go2rtc configuration** — set up as part of every Pi deployment for proxying CP Plus RTSP stream to dashboard via WebRTC
- **Camera IP/DHCP fix** — camera IP changed from 192.168.50.102 to .100 (DHCP moved it); stream name fixed to match MongoDB record
- **Multi-camera support (v3)** — CAMERAS array replaces single CAMERA_IP/CAMERA_STREAM_NAME; go2rtc.yaml generated with ALL camera streams; camera-wrapper TCP probes all cameras in parallel

- **Bulk heartbeat (v3)** — camera-wrapper calls /api/camera/bulk-heartbeat with array of all camera statuses instead of single-camera heartbeat

8. Raspberry Pi Hub — Infrastructure (22)

Central hub running Zigbee2MQTT, MQTT broker, go2rtc, dashboard, and all bridges.

- **Multiple pisetup versions built:** v3, v4, v5, v8, v9, v20, v23, v25, v26, v27, v27.1, v27.2, v28 — each adding fixes and features
- **Fresh Pi setup completed** — full deploy: dashboard + camera + Zigbee + MQTT + GLK + hub heartbeat
- **Home mapping** — POST /api/hub/setup with home_id to bind Pi to a specific home
- **Zigbee service + cron checkers fixed for home-based device mapping** — checkers were looking up devices by resident instead of by home
- **DS3231 RTC clock chip setup on QA Pi** — hardware real-time clock for accurate timestamps without NTP
- **Hub heartbeat implementation** — periodic heartbeat from Pi to cloud backend so app knows hub is online
- **SD card corruption root cause analysis** — deep investigation identifying SDIO bus (shared between WiFi and SD card) as hardware-level root cause; formal DOCX report produced
- **SSD migration (USB boot)** — EEPROM bootloader update, Source adapter compatibility (UAS quirks, autosuspend disable, blacklist), cmdline.txt + config.txt config, live filesystem auto-expansion
- **Hub status log collection** — created hub_status_log MongoDB collection with model + API endpoint
- **Hardware watchdog (v28)** — enabled Pi's hardware watchdog timer for auto-reboot on hang
- **Network hardening (v28)** — WiFi reconnect scripts, connection monitoring
- **Cudy TR1200 router internet connectivity fix** — diagnosed and fixed QA network internet access
- **Autossh tunnel replaced by Cloudflare named tunnel (v5)** — token-based named tunnel; installs cloudflared, registers URL via POST /api/hub/tunnel
- **Cloudflare ingress rules fix (v23)** — token-based cloudflared sometimes fails to pull ingress config causing 'No ingress rules' + HTTP 503; fix creates explicit config.yml
- **Mosquitto MQTT broker + Zigbee2MQTT** — configured on each Pi deployment with SONOFF Zigbee dongle
- **Auto-retry function (v23)** — retry() with exponential backoff (5s-60s, 6 attempts) applied to apt-get, Node.js install, npm install, git clone/pull, cloudflared download
- **Service health checks (v23)** — check_service_health() verifies systemd services are active after start, auto-restarts up to 3 times; check_pm2_health() for PM2 processes
- **PM2 boot persistence fix** — Pi was dead after 12hr power-off; pm2 startup systemd now runs unconditionally; falls back to manual systemd service creation; @reboot cron as backup
- **Static IP baked into setup script (v26)** — static IP 192.168.50.106 configured inside setup-pi-fresh.sh; supports both dhcpcd (Bullseye) and NetworkManager (Bookworm+)
- **Live filesystem auto-expansion (v27.2)** — auto-detects undersized root (<10GB free) and expands live using parted + resize2fs without reboot
- **Deploy script gateway exclusion (v5)** — Cudy router excluded from Pi auto-discovery because its Broadcom chip responded to SSH probe
- **Deploy scripts for Mac + Windows** — deploy-pi.command (Mac, sshpass) and deploy-pi.ps1 (Windows, Posh-SSH) with env selector (Prod/QA)

9. Cross-Device / Platform-Wide (20)

Fixes and features spanning multiple devices or the platform as a whole.

- **Resident dropdown removal from ALL device forms except GLK** — simplified pairing flow; home-based mapping replaced per-resident assignment
- **Sync button implementation** — dashboard UI button to force-sync all device states
- **handleSave fix** — frontend re-fetches after save — after saving device config, UI pulls fresh data instead of showing stale state
- **Device listing ignoring home filter — fixed** — device list API wasn't filtering by selected home
- **"Unauthorized device access" in device_controller.js — fixed** — legitimate device requests were being rejected
- **Alerts route ordering in user_device_route.js — fixed** — route parameter conflicts caused alerts endpoint to 404

- **debug_dashboard_data diagnostic function added** — helper to dump full dashboard data state for troubleshooting
- **Timezone bug in requested_date — fixed** — IST vs UTC offset was shifting which day's data was returned
- **ESLint snake_case enforcement** — applied id-match rule across codebase
- **CI/CD git pull failure diagnosed and fixed** — deployment pipeline was broken
- **EC2 Docker container name conflict resolved** — QA backend container couldn't start due to name collision
- **server.js corruption fix** — removed duplicated content + merge marker from server.js after bad merge
- **All merge conflicts resolved on both repos** — qa + master branches for both Pi Dashboard and Backend repos
- **Push notification event/screen mapping** — wired up which alert types trigger which push notifications and which app screen they deep-link to
- **Zigbee type auto-detection fix (v25)** — mqttClient.js type detection broken for freshly-paired devices; also checks dev.definition.exposes array; DeviceForm.jsx got Temperature, Leak, Zigbee added to type dropdown
- **MQTT bridge dotenv fix (v4)** — bridge always hitting production because dotenv wasn't imported before reading process.env.BACKEND_URL; added inline env vars via PM2 as backup
- **Frontend auth.js fix (v4)** — login was getting prod token which Pi forwarded to QA causing 401 Unauthorized; fixed auth.js to always read VITE_BACKEND_URL; .env created for both environments
- **server.js assign-name crash fix (v4)** — Z2M configuration.yaml may have no 'devices:' key on fresh installs; original code crashed with TypeError; fixed with try/catch + null check
- **Diagnosed dashboard empty data for Sep 17** — investigated and explained why dashboard showed no data for a specific date
- **Implementation spec documents created** — formal specification documents for device integration and dashboard logic